

MODULE 03 · DOMAIN 2 - GENERATIVE AI FUNDAMENTALS

Generative AI Fundamentals

How foundation models actually work, and the vocabulary the exam tests.

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Traditional ML predicts **a label or a number.**
Generative AI produces **new content.**

What makes a model a foundation model

- ◆ Trained on **enormous, broad, mostly unlabelled** data. Not one task, not one domain.
- ◆ **Self-supervised** — the data supplies its own labels by hiding parts of itself.
- ◆ **Adaptable** — one model serves summarisation, classification, translation, code.
- ◆ An **LLM** is a foundation model specialised to language. All LLMs are FMs; not all FMs are LLMs.

ON THE EXAM

The distinguishing feature is breadth. A model trained for one task is not a foundation model, however large it is.

What happens to your prompt

01

Tokenise

Text is split into tokens: words, word fragments, punctuation.

02

Embed

Each token becomes a vector capturing its meaning in context.

03

Attend

Self-attention weighs how much each token matters to every other.

04

Predict

The model outputs a probability for every possible next token.

05

Sample

One token is chosen. Then the whole thing repeats.

ON THE EXAM

Generation is one token at a time. Everything you can control at inference — temperature, top-p, length — acts on the sample step.

The unit of everything

- ◆ A token is roughly **three quarters of an English word**. 100 tokens is about 75 words.
- ◆ You are billed per token, **input and output separately**. Cost questions are token questions.
- ◆ **Different models tokenise differently**. The same sentence is not the same token count everywhere.
- ◆ Rare words, other languages and code all consume more tokens per character.

ON THE EXAM

There is a question asking why the same text produces different token counts across models. The answer is that each model has its own tokenizer.

Meaning as coordinates

- ◆ An embedding is a **vector** — a list of numbers representing meaning.
- ◆ Similar meanings land close together, so **distance measures similarity**.
- ◆ This is what makes semantic search and RAG retrieval possible.
- ◆ Query and documents must use the **same embedding model**, or the vectors are not comparable.

ON THE EXAM

Embeddings turn a meaning problem into a geometry problem. That single idea underpins vector databases and retrieval.

Everything the model can see at once

- ◆ The context window is the **maximum tokens in a single request**: prompt plus response.
- ◆ The model has **no memory between calls**. Conversation history is resent every time.
- ◆ Bigger windows cost more per call and usually run slower.
- ◆ Exceed it and the request fails or the oldest content is dropped.

ON THE EXAM

'The model forgot what we discussed' is a context window question, not a training question. Nothing is remembered between calls.

Four stages, and the exam tests the difference

01

Pre-training

Enormous general corpus, self-supervised. Months and millions of dollars.

02

Continued pre-training

More unlabelled data, but domain-specific. Teaches vocabulary.

03

Fine-tuning

Labelled examples of the task. Teaches behaviour and format.

04

RLHF

Human preferences as reward. Aligns tone, safety and helpfulness.

ON THE EXAM

You will not do pre-training. You might do the other three. The exam mostly tests which of stages two and three fits a described need.

Continued pre-training or fine-tuning?

CONTINUED PRE-TRAINING

Teach it new vocabulary

- Input is unlabelled domain text
- The model does not know your field's language
- Medical, legal or scientific terminology
- Feed it documents, not question-answer pairs
- Broad domain knowledge, not a specific task

FINE-TUNING

Teach it a behaviour

- Input is labelled examples of the task
- The model understands the words but responds wrongly
- Specific tone, format or structure required
- Prompt and ideal completion pairs
- One task done well, not general knowledge

THE TIEBREAKER

Follow the data. Unlabelled documents means continued pre-training. Labelled prompt-and-response pairs means fine-tuning.

How a model learns to behave

- ◆ **Instruction fine-tuning** trains on prompt-and-response pairs so the model follows instructions.
- ◆ **RLHF** has humans rank outputs. Those rankings train a reward model.
- ◆ The reward model then guides the FM towards responses humans prefer.
- ◆ This is how alignment with company values and tone is achieved.

ON THE EXAM

Any scenario about aligning a model to human preferences, values or acceptable tone is pointing at RLHF.

The idea underneath all customisation

- ◆ Take a model trained on one task and reuse its learned representations for another.
- ◆ Far cheaper than training from scratch, and needs far less data.
- ◆ Fine-tuning is transfer learning applied to a foundation model.
- ◆ Also the reason a vision model trained on general images adapts to your products.

ON THE EXAM

If an option describes reusing a pre-trained model's knowledge for a new but related task, that is transfer learning.

Three families, three jobs

ARCHITECTURE	WHAT IT IS FOR
Transformer	Text and sequences. Self-attention. Behind every LLM.
Diffusion	Images. Starts from noise and removes it step by step.
GAN	Images. A generator and a discriminator competing against each other.

ON THE EXAM

Transformer means language. Diffusion means image generation, and is what Stable Diffusion and Nova Canvas use.

Why self-attention mattered

- ◆ **Self-attention** lets every token weigh every other token directly.
- ◆ Long-range meaning survives. 'It' can be linked to a noun forty words earlier.
- ◆ Processing is **parallel**, unlike older sequential RNNs. That is what enabled the scale.
- ◆ Introduced in the 2017 paper 'Attention Is All You Need'.

More than one kind of input

- ◆ Accept combinations of text, image, video or audio in a single request.
- ◆ Describing an image, answering questions about a chart, summarising a video.
- ◆ Nova Lite and Nova Pro take text, image and video in. All of them output text.
- ◆ Multimodal **input** is not the same as multimodal **output**. Check which the question means.

ON THE EXAM

A model that reads images and writes text is multimodal. Generating an image needs Nova Canvas or Stable Diffusion instead.

The Nova family

MODEL	SHAPE
Nova Micro	Text only. Lowest latency and cost. 128K context.
Nova Lite	Multimodal in, text out. Fast and cheap. 300K context.
Nova Pro	Multimodal. Best balance of accuracy, speed and cost. 300K.
Nova Premier	Most capable. 1M context. Teacher model for distillation.
Nova Canvas	Image generation and editing.
Nova Reel	Short video generation.

ON THE EXAM

Text-only and cheapest is Micro. Needs to see an image is Lite or Pro. Needs to make an image is Canvas.

You are not limited to Amazon's models

- ◆ **Amazon Titan** — Amazon's earlier family. Text, image, and the embeddings models.
- ◆ **Anthropic Claude, Meta Llama, Mistral, Cohere, AI21, Stability AI.**
- ◆ All reached through the **same Bedrock API**. Swapping models is a parameter change.
- ◆ That uniformity is a stated advantage of Bedrock, and the exam expects you to know it.

ON THE EXAM

Questions about comparing or switching between models are really about Bedrock giving one API across many providers.

What generative AI is genuinely good at

- ◆ **Summarising** long documents, calls or threads into short form.
- ◆ **Drafting and rewriting** — marketing copy, emails, documentation, tone changes.
- ◆ **Code** — generating, explaining, and translating between languages.
- ◆ **Classification and extraction** from unstructured text, often with no training data.
- ◆ **Conversation** — support agents, internal assistants, question answering.

ON THE EXAM

Converting code from one language to another is a generative task. It shows up as a question and the answer is a foundation model, not a traditional ML service.

What it is bad at, and the exam knows

- ◆ **Factual reliability.** Fluent and wrong is the characteristic failure.
- ◆ **Maths and precise calculation.** It predicts plausible tokens, it does not compute.
- ◆ **Knowledge cutoff.** Nothing after training, unless you supply it in the prompt.
- ◆ **Nondeterminism.** The same prompt can return different answers.
- ◆ **Cost and latency** scale with tokens, which surprises teams at production volume.

ON THE EXAM

When a scenario needs a guaranteed, identical, auditable answer every time, generative AI is the wrong tool. Say so.

The most-tested failure mode

- ◆ Output that is **fluent, confident and false**. Invented citations, fake product details.
- ◆ It is not a bug being fixed. It follows from predicting likely tokens.
- ◆ **RAG** is the standard mitigation — ground answers in retrieved source material.
- ◆ Also: lower temperature, explicit prompt constraints, and human review.
- ◆ Fine-tuning reduces it in a domain but never removes it.

ON THE EXAM

Hallucination appears repeatedly across the exam. When a question asks how to improve factual accuracy, RAG is the default correct answer.

Why the same prompt gives different answers

- ◆ The model produces a **probability distribution**, then samples from it.
- ◆ Sampling introduces variation, which is what makes output feel creative.
- ◆ **Lower temperature** makes it more focused and repeatable.
- ◆ Even at temperature zero, identical output across runs is not guaranteed.

ON THE EXAM

A question asking why responses differ between identical requests is testing whether you know generation samples rather than looks up.

Carry these into the exam

- ◆ A token is about three quarters of a word, and every model tokenises differently.
- ◆ Context window covers prompt plus response. No memory exists between calls.
- ◆ Unlabelled domain documents means continued pre-training. Labelled pairs means fine-tuning.
- ◆ Aligning to human preferences means RLHF.
- ◆ Transformer for language, diffusion for images.
- ◆ Hallucination is answered by RAG. Nondeterminism is answered by temperature.

A law firm finds that a foundation model misreads its contracts. It does not recognise terms like novation, estoppel or subrogation, and treats them as ordinary words.

The firm has forty years of contracts and legal memoranda, but no question-and-answer datasets.

Which customisation approach fits?

- ☐ A Instruction fine-tuning
- ☐ B Continued pre-training
- ☐ C RLHF
- ☐ D Train a foundation model from scratch

◆ PAUSE AND DECIDE

Which customisation approach fits?

- A

 Instruction fine-tuning
Needs labelled pairs. The firm has documents, not pairs.
- B

 Continued pre-training
Unlabelled domain text teaching vocabulary. Exactly this.
- C

 RLHF
Aligns tone and preference. The problem is vocabulary.
- D

 Train from scratch
Months and millions, to solve a vocabulary gap.

ON THE EXAM

Two clues, both decisive: the failure is vocabulary, and the available data is unlabelled documents.

A retailer builds an assistant on a foundation model. Testing shows it confidently invents product specifications and delivery policies that do not exist.

The product catalogue and policy documents are accurate, current, and stored in S3.

What is the most effective fix?

- ☐ A Increase the temperature
- ☐ B Implement RAG over the catalogue and policies
- ☐ C Move to a model with a larger context window
- ☐ D Apply RLHF

◆ PAUSE AND DECIDE

What is the most effective fix?

- A

 Increase the temperature
Backwards. Higher temperature increases invention.
- B

 Implement RAG
Grounds answers in the accurate documents they already have.
- C

 Larger context window
More room for text the model still does not have.
- D

 Apply RLHF
Shapes tone and preference. Does not supply missing facts.

ON THE EXAM

Confident invention plus accurate documents already sitting in S3 is the RAG signature. The documents being mentioned is the giveaway.

A company runs the same summarisation prompt through a foundation model twice and gets two different summaries. Both are accurate.

Compliance requires that an audit rerun produces the identical output.

What should the team understand and do?

- ☐ A The model is faulty; raise a support case
- ☐ B Set temperature to 0 and accept it is still not guaranteed
- ☐ C Fine-tune the model to make it deterministic
- ☐ D Increase the context window

◆ PAUSE AND DECIDE

What should the team understand and do?

- A

 Model is faulty
It is behaving correctly. Generation samples by design.
- B

 Temperature 0, still not guaranteed
The honest answer. Reduces variation without promising none.
- C

 Fine-tune for determinism
Fine-tuning changes behaviour, not the sampling mechanism.
- D

 Larger context window
Unrelated to output variability.

ON THE EXAM

Watch for options that promise a guarantee generative AI cannot give. The correct answer is often the one that admits a limit.

A media company wants an application where a user uploads a photograph of a room and asks questions about it in text, such as what style the furniture is.

The response should be a written description. No images need to be created.

Which kind of model does this require?

- ☐ A A text-only model such as Nova Micro
- ☐ B An image generation model such as Nova Canvas
- ☐ C A multimodal model such as Nova Pro
- ☐ D An embeddings model such as Titan Embeddings

◆ PAUSE AND DECIDE

Which kind of model does this require?

- A

 Nova Micro
Text only. Cannot accept an image as input.
- B

 Nova Canvas
Generates images. The scenario says none are needed.
- C

 Nova Pro
Accepts image plus text in, returns text out.
- D

 Titan Embeddings
Produces vectors for retrieval, not written answers.

ON THE EXAM

Check input and output separately. Image in, text out is multimodal input — not image generation.